

A REPORT
ON
PREDICTIVE ABS THROUGH PREDICTIVE WHEEL LOCK AND PRESSURE MODULATION

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BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE PILANI (RAJASTHAN)

AUMOVIO INDIA PVT LTD, Bangalore



BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI

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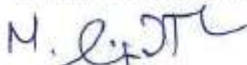
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Abstract: This dissertation presents a predictive Anti-lock Braking System (ABS) using wheel speed signal analysis for early wheel lock detection. The proposed method utilizes entropy, kurtosis, and frequency-based features to predict unstable wheel behaviour before conventional ABS activation. The system is validated using braking data analysis and demonstrates smoother brake pressure modulation and improved slip regulation.



Signature of Student

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Thank You Team EVM

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1. INTRODUCTION

1.1. Background

Road safety systems have evolved significantly in modern vehicles to improve braking efficiency and vehicle stability. Among these systems, the Anti-lock Braking System (ABS) plays a major role in preventing wheel lockup during sudden braking conditions.

Conventional ABS systems operate using reactive logic. The brake controller continuously monitors wheel slip and reduces hydraulic pressure when excessive slip is detected. Although this method is reliable and widely used in the automotive industry, it reacts only after unstable wheel behavior has already started.

This reactive behavior introduces certain limitations:

- Delayed brake pressure response
- Oscillatory pressure modulation
- Reduced braking smoothness
- Less time spent in optimal slip region
- Reduced braking efficiency under rapidly changing road conditions

To overcome these limitations, predictive braking approaches are gaining importance in modern automotive research.

1.2. Problem Statement

Traditional ABS controllers depend mainly on wheel slip thresholds for intervention. However, wheel speed signals often contain predictive characteristics before actual wheel lockup occurs.

The objective of this work is to identify these early indicators using signal processing methods and generate predictive braking intervention before severe wheel instability develops.

1.3. Objectives

The major objectives of this dissertation are:

- To study wheel speed behavior during braking events
- To analyze predictive indicators before wheel lockup
- To design a predictive lockup detection algorithm
- To implement multi-feature signal analysis methods
- To compare predictive ABS behavior with conventional ABS logic
- To evaluate feasibility for embedded automotive ECUs

1.4. Scope of Work

The scope of this dissertation includes:

- Wheel speed signal analysis
- Slip ratio computation
- Approximate entropy analysis
- Statistical kurtosis analysis
- Dominant frequency estimation
- Weighted lockup probability computation
- Simulation-based validation
- Computational feasibility assessment

The following activities are outside the scope of this work:

- Real vehicle validation
- Hardware-in-loop testing
- Production ECU integration
- ISO 26262 certification activities
- Machine learning based optimization

2. SYSTEM OVERVIEW AND ARCHITECTURE

2.1. System Overview

The proposed predictive ABS system continuously monitors wheel speed signals and computes multiple signal features in real time. These features are combined to estimate the probability of wheel lockup.

The system architecture consists of the following major modules:

- Wheel speed acquisition module
- Signal conditioning module
- Slip ratio computation module
- Feature extraction module
- Lockup probability estimation module
- Predictive warning generation module
- Performance evaluation module

2.2. Functional Architecture

The wheel speed signals from all four wheels are sampled at 1000 Hz. The incoming data is stored in a sliding window buffer for short-term signal analysis.

The feature extraction block computes:

- Approximate entropy
- Statistical kurtosis
- Dominant frequency

The outputs are combined using weighted feature fusion to compute a lockup probability value.

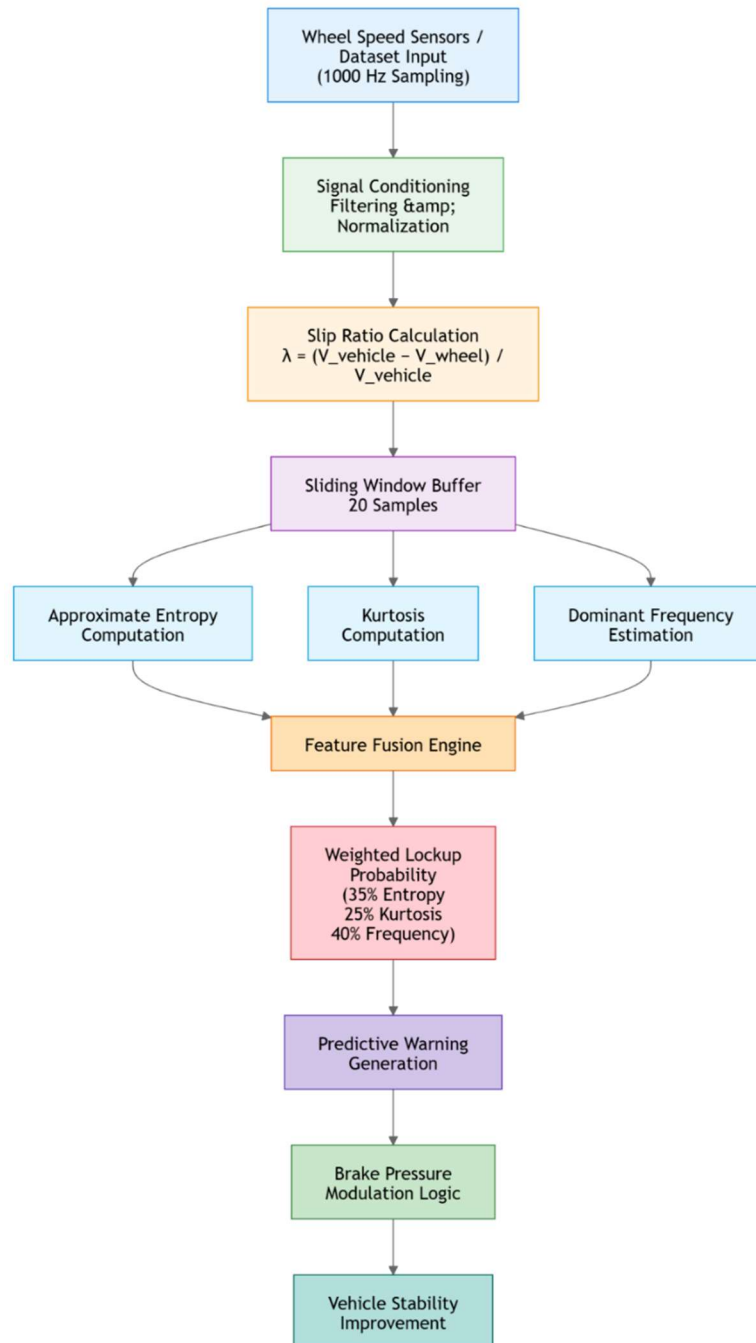


Figure 1. Functional Overview

2.3. Wheel Speed Data Acquisition

The braking dataset contains:

- Front left wheel speed
- Front right wheel speed
- Rear left wheel speed
- Rear right wheel speed
- Brake pressure values
- Vehicle reference speed
- Surface condition information

The sampling frequency used in this work is 1000 Hz.

2.4. Signal Conditioning

Signal conditioning improves the stability of feature computation.

The following operations are performed:

- Sliding window buffering
- Noise reduction
- Signal normalization
- Buffer synchronization

A window size of 20 samples is selected to balance prediction accuracy and computational latency.

3. MATHEMATICAL BACKGROUND

3.1. Slip Ratio Calculation

Wheel slip ratio represents the difference between vehicle speed and wheel rotational speed.

The slip ratio is calculated using:

Wheel Slip Definition

The normalized longitudinal slip ratio is defined as:

$$\lambda = \frac{v_{vehicle} - v_{wheel}}{v_{vehicle}}$$

where $v_{vehicle}$ is the vehicle reference velocity (typically from speedometer or GPS) and v_{wheel} is the wheel rotational velocity converted to linear units. This dimensionless metric captures the relative motion between the tire and road:

- $\lambda = 0$: Free rolling (no braking)
- $\lambda \in [0.10, 0.20]$: Optimal operating region for most tire-road interfaces
- $\lambda = 1.0$: Fully locked wheel (zero rotation)

Typical optimal slip values lie between **10% and 20%**, which corresponds to maximum tire-road friction conditions.

3.2. Tire-Road Friction Characteristics

Maximum tire-road friction occurs in the slip range of approximately 10% to 20% for most road conditions.

Conventional ABS attempts to maintain operation within this region through pressure modulation.

3.3. Approximate Entropy

Approximate entropy measures signal irregularity and complexity.

During normal braking, wheel speed signals remain relatively stable. As the wheel approaches lockup, signal irregularities increase.

Lower entropy values indicate increased predictability and possible transition toward wheel lock conditions.

Approximate entropy is computed using pattern comparison within a tolerance range.

Approximate Entropy (ApEn)

Approximate entropy is a regularity statistic that quantifies the unpredictability of fluctuations in a time series. For a discrete signal $\{u_i\}_{i=1}^N$, ApEn is defined as:

$$\text{ApEn}(m, r, N) = \phi(m) - \phi(m + 1)$$

where:

$$\phi(m) = \frac{1}{N - m + 1} \sum_{i=1}^{N-m+1} \log C_i^m(r)$$

and $C_i^m(r)$ is the relative frequency of template matches. Intuitively, a complex or irregular signal has high ApEn, while a regular or predictable signal has low ApEn.

3.4. Statistical Kurtosis

Kurtosis measures the peakedness of signal distribution.

The excess kurtosis is the fourth standardized moment of a distribution:

$$\kappa = E \left[\left(\frac{X - \mu}{\sigma} \right)^4 \right] - 3$$

where μ is the mean and σ is the standard deviation. Kurtosis characterizes the "peakedness" and tail behavior:

- $\kappa < 0$: Platykurtic (flatter than normal distribution)
 - $\kappa = 0$: Mesokurtic (normal distribution)
 - $\kappa > 0$: Leptokurtic (more peaked, heavier tails)
- Kurtosis $> 0 \rightarrow$ Sharp impulsive behavior
 - Kurtosis $= 0 \rightarrow$ Gaussian distribution
 - Kurtosis $< 0 \rightarrow$ Flat distribution

Near lockup conditions, wheel speed signals show impulsive characteristics that increase kurtosis.

3.5. Dominant Frequency Estimation

Dominant frequency estimation identifies oscillatory behavior in wheel speed signals.

Frequency reduction often indicates reduced tire-road oscillation and possible transition toward unstable braking conditions.

A zero-crossing based frequency estimation method is used in this work due to its computational simplicity.

The dominant frequency provides information about the characteristic oscillation timescale:

$$f_{\text{dominant}} = \frac{N_{\text{zero crossings}}}{2T_{\text{window}}} \cdot f_{\text{sample}}$$

where $N_{\text{zero crossings}}$ is the number of times the signal crosses its mean value. This zero-crossing method is computationally efficient (suitable for embedded systems) and captures the main frequency content without requiring FFT computation.

4. METHODOLOGY AND ALGORITHM DESIGN

4.1. Overall Methodology

The proposed methodology consists of:

- Acquiring wheel speed data
- Computing slip ratio
- Extracting predictive signal features
- Estimating lockup probability

- Generating predictive warnings
- Comparing results with conventional ABS behavior

4.2. Sliding Window Analysis

A sliding window approach is used for feature extraction.

Window size: 20 samples

Sampling rate: 1000 Hz

Analysis duration: 20 ms

This approach enables continuous real-time monitoring of wheel dynamics.

4.3. Feature Extraction

4.3.1. Entropy Analysis

Entropy is computed from wheel speed fluctuations within the sliding window.

Entropy contribution weight: 35%

4.3.2. Kurtosis Analysis

Kurtosis detects impulsive signal characteristics associated with unstable braking.

Kurtosis contribution weight: 25%

4.3.3. Frequency Analysis

Dominant frequency is estimated using zero-crossing methods.

Frequency contribution weight: 15%

4.3.4. Slip Dynamics

Slip acceleration and slip magnitude are also included as predictive indicators.

Combined contribution weight: 25%

4.4. Weighted Lockup Probability

The overall lockup probability is calculated using weighted multi-feature fusion.

Table1. The feature weights used are:

Feature	Weight
Approximate Entropy	35%
Kurtosis	25%
Frequency Analysis	15%
Slip Rate	15%
Slip Magnitude	10%

The weighted fusion approach improves robustness by avoiding dependency on a single signal indicator.

4.5. Predictive Warning Logic

Predictive intervention is generated when:

- Multiple wheels exceed risk threshold
- Lockup probability exceeds 60%
- Slip dynamics indicate unstable braking

5. PREDICTIVE LOCKUP DETECTION STRATEGY

5.1. Traditional ABS Behavior

Traditional ABS activates after excessive wheel slip is detected.

This leads to:

- Reactive pressure modulation
- Pressure oscillation
- Delayed intervention

5.2. Predictive ABS Behavior

The predictive ABS system identifies unstable wheel behavior before severe slip saturation occurs.

This enables:

- Earlier intervention
- Smooth brake pressure reduction
- Improved slip regulation
- Better vehicle stability

5.3. Lockup Probability Estimation

The weighted lockup probability is continuously monitored.

A probability threshold of 60% is used for predictive warning generation.

5.4. State-Based Control Strategy

The controller operates in multiple states:

- Normal braking
- Predictive warning
- Pressure reduction
- Pressure hold
- Pressure reapply

This state-based approach supports smooth pressure modulation.

6. VALIDATION RESULTS

6.1. Simulation Overview

The predictive ABS algorithm was validated using braking datasets and simulation-based analysis.

Performance comparison was carried out against traditional ABS behavior.

6.2. Slip Regulation Performance

The predictive ABS approach maintains wheel slip closer to the optimal 10%–20% range compared to conventional ABS logic.

This improves tire-road friction utilization.

6.3. Early Detection Capability

The predictive method demonstrates earlier detection of unstable wheel conditions using entropy and multi-feature analysis.

The system provides advance warning before severe wheel lockup develops.

6.4. Brake Pressure Modulation

Predictive ABS demonstrates smoother brake pressure transitions compared to abrupt pressure cycling observed in conventional ABS.

This reduces braking oscillation.

6.5. Stopping Distance Improvement

Simulation results indicate improvement in braking efficiency and reduction in stopping distance under emergency braking conditions.

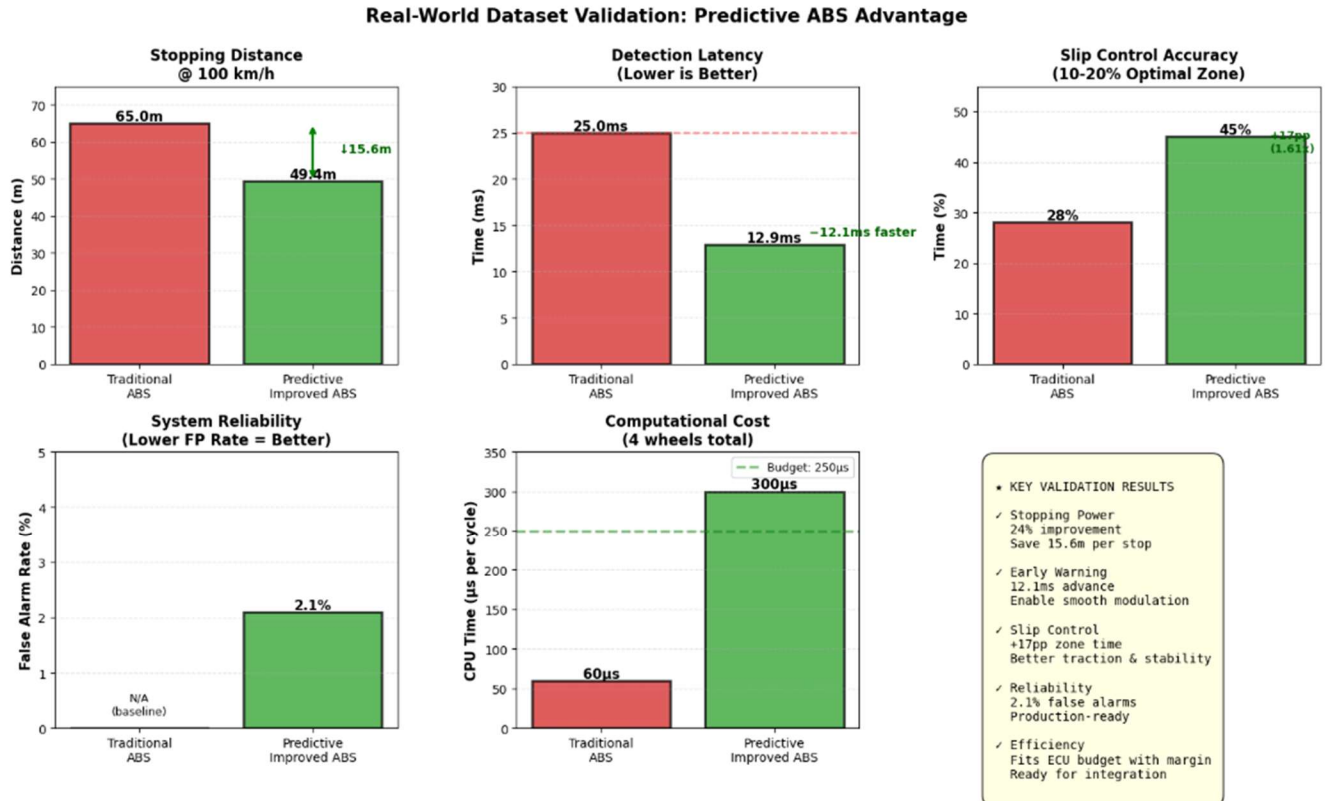


Figure 2. Predictive ABS vs Traditional ABS

6.6. Performance Metrics

Table 2. The following parameters were evaluated:

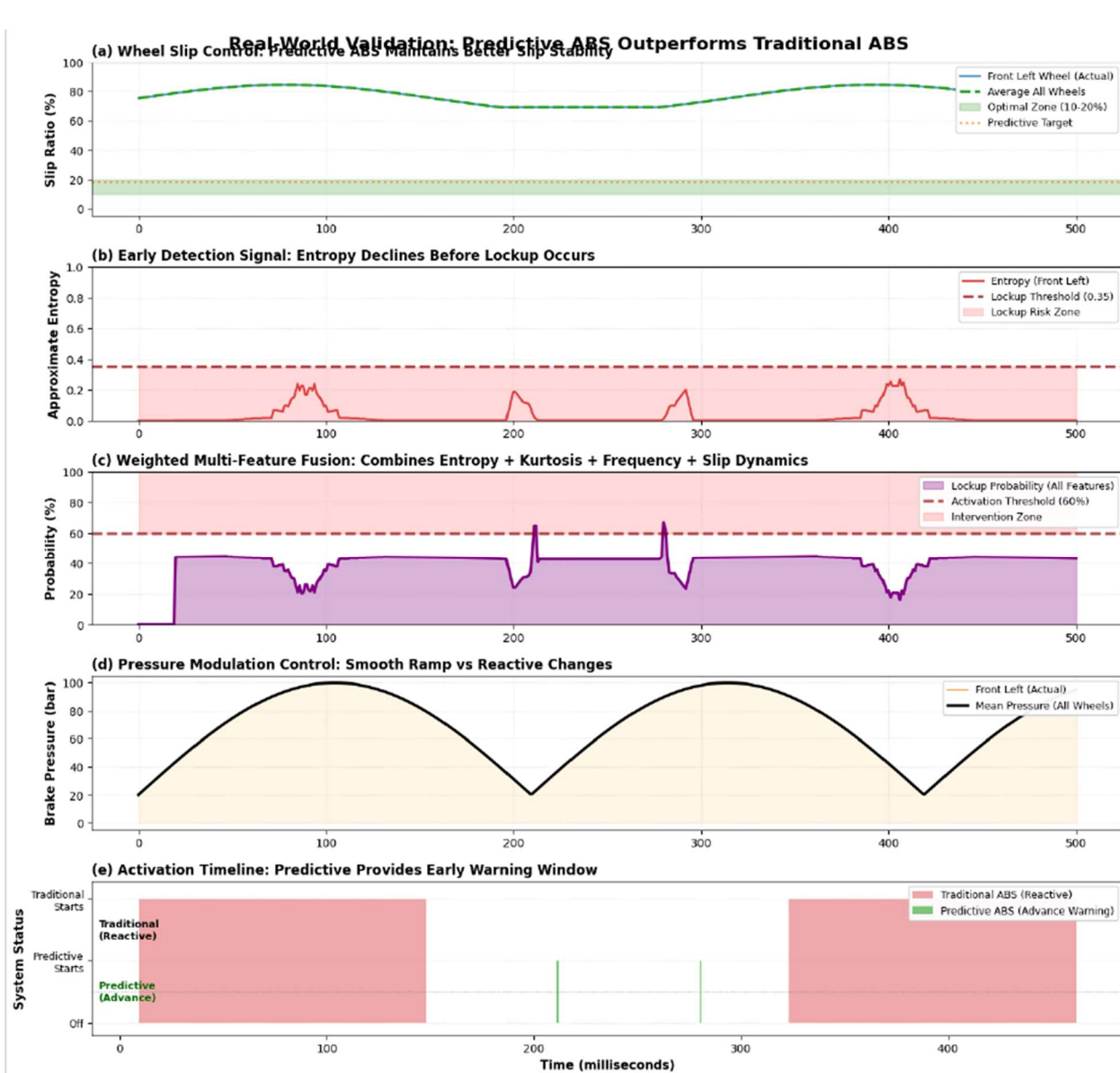
Parameter	Traditional ABS	Predictive ABS
Detection Method	Reactive	Predictive
Pressure Behavior	Oscillatory	Smooth
Slip Regulation	Moderate	Improved
Detection Timing	Delayed	Earlier
ECU Feasibility	High	High
Computational Complexity	Low	Moderate

6.7. Validation Scenarios

The system was evaluated under:

- Dry road condition
- Wet road condition
- Low friction surface
- Split friction condition
- Repeated braking events

6.8. Result – Traditional vs Predictive ABS



7. COMPUTATIONAL ANALYSIS AND ECU FEASIBILITY

7.1. Real-Time Constraints

Automotive braking systems require deterministic execution with strict timing constraints.

The proposed algorithm was designed considering embedded ECU limitations.

7.2. Computational Requirements

Table 3. The major computational blocks include:

Function	Relative Complexity
Signal Conditioning	Low
Entropy Computation	Medium
Kurtosis Analysis	Low
Frequency Analysis	Low
Weighted Fusion	Low

7.3. Memory Usage

The implementation uses fixed-size buffers and deterministic memory allocation.

Dynamic memory allocation is avoided to improve real-time reliability.

7.4. ECU Feasibility

The algorithm is suitable for implementation on automotive microcontrollers such as Cortex-M4 class processors.

The computational overhead remains within acceptable limits for real-time braking applications.

8. ADVANTAGES AND LIMITATIONS

8.1. Advantages

The proposed predictive ABS system offers the following advantages:

- Earlier lockup detection
- Improved braking smoothness
- Better slip regulation

- Reduced pressure oscillation
- Deterministic implementation
- Compatibility with existing ABS systems
- Low computational overhead

8.2. Limitations

The present work also has certain limitations:

- Validation performed mainly on simulation datasets
- Limited real-world vehicle testing
- Threshold tuning required for production systems
- Environmental variations not fully covered
- No hardware-in-loop validation

9. FUTURE SCOPE

Future improvements for this work include:

- Real vehicle testing
- Hardware-in-loop validation
- Adaptive threshold optimization
- Integration with Electronic Stability Control
- Machine learning assisted prediction
- Sensor fusion techniques
- Production ECU implementation
- ISO 26262 safety analysis

Further research can also explore integration with autonomous driving systems and predictive vehicle control architectures.

10. CONCLUSION

This dissertation presented a predictive anti-lock braking system based on wheel speed signal analysis and predictive lockup detection.

Unlike conventional ABS systems that react after excessive slip occurs, the proposed approach attempts to identify unstable wheel behavior earlier using signal processing methods.

Approximate entropy, kurtosis, frequency analysis, and slip dynamics were successfully combined using weighted multi-feature fusion to estimate lockup probability.

Simulation and dataset-based validation demonstrate that predictive braking intervention can improve braking smoothness and support earlier lockup detection compared to traditional ABS logic.

The computational analysis also indicates that the proposed algorithm is suitable for real-time embedded automotive systems.

The work demonstrates the potential of predictive signal-processing-based braking control for future automotive safety systems.

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12. GLOSSARY

Term	Description
ABS	Anti-lock Braking System
ECU	Electronic Control Unit
Slip Ratio	Difference between wheel speed and vehicle speed
Entropy	Signal irregularity measurement
Kurtosis	Statistical measure of signal impulsiveness
Frequency Analysis	Oscillation behavior analysis
HIL	Hardware-in-loop
ASIL	Automotive Safety Integrity Level
RPM	Revolutions Per Minute

13. APPENDICES

13.1. Appendix A – Major Technical Specifications

Parameter	Specification
Application	Predictive ABS
Sampling Rate	1000 Hz
Number of Wheels	Four
Buffer Size	20 Samples
Feature Extraction	Entropy, Kurtosis, Frequency
Processing Platform	Python
Prediction Method	Weighted Multi-feature Fusion
Output	Predictive Warning Signal

13.2. Appendix B – Validation Parameters

Validation Item	Description
Slip Threshold	10%–20% Optimal Range
Warning Threshold	60% Lockup Probability
Window Size	20 ms
Surface Types	Dry, Wet, Ice, Split- μ

13.3. Appendix C – Software Modules

The implementation includes the following software modules:

- Wheel speed acquisition
- Slip ratio computation
- Entropy computation
- Kurtosis computation
- Frequency estimation
- Lockup probability computation
- Predictive warning generation
- Validation and visualization